Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Terrabotics
Project / Solution Name	SimpanAI
Name (Leader)	Lo Yu Ao
Email & Phone (Leader)	loaidan0608@gmail.com +60195310530
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://SimpanAI.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/Yang060429/SDCHackathon_Terrabotics

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Seller interface: The seller uses React and TypeScript as frontend that is built with Vite. It handles inventory, barcode check-in, sales history, forecasting, resupply, and stock history.
Frontend to API: The UI uses TanStack Query and a generated OpenAPI client to send REST requests to the /api routes. The browser sends the session cookie with those requests.
Backend logic: A Node.js and TypeScript API built with Express validates requests, checks the seller's session and data ownership, and applies inventory and order rules that include partial deliveries and FIFO batch handling.
Database: The API reads and writes PostgreSQL through Drizzle ORM and node-postgres. The database stores account, product, sales, order, batch, and movement data. The code connects using DATABASE_URL.
Forecasting: For the forecast, the API reads product and dated sales data, then starts a local Python process and passes it JSON through standard input. The Python process returns forecast values to the API, which calculates suggested restock quantities and returns the results to the UI.
AI/model status: This is not a call to a separate hosted AI API. The project has PyTorch and LightGBM model files, the current code uses its statistical fallback, based on recent sales, trend, and weekday patterns.
Hosting: The project configuration defines a Replit-hosted static frontend and a separate Node/Express API service. The Python forecasting process runs alongside the API. The database's hosting location is not specified.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.3	Main language for the frontend and backend application code	Apache License 2.0
React 19.1.0	Build the seller interface	MIT
Vite 7.3.7	Run and build the frontend	MIT

Name & Version	Purpose in Project	License
Node.js; Express 5.3.0	Runs the backend and provides the REST API	MIT
PostgreSQL; Drizzle ORM 0.45.4;node-postgres 8.23.1	Relational data storage and database access	PostgreSQL License; Apache 2.0; MIT, respectively
Python; PyTorch 2.6;; LightGBM 4.5; Numpy 1.26; pandas 2.2; SciPy 1.12	Forecasting process and optional trained-model inference	PSF License; BSD-3-Clause for PyTorch, Numpy, pandas, and SciPy; MIT for LightGBM

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Fronts CSS API/CDN	Loads the Seller Center's web front, including Plus Jakarta Sans and JetBrains Mono.	free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit-Seller Centre web service	Host the React/Vite fronted. Its production configuration builds and serves static files	Replit preview is available. Whether it has been published, and the plan and region, are not shown in the project file.
Replit-API Server service	Run the Node.js/ Express backend under /api ; the local Python forecasting process runs alongside it.	Production run configuration exists. Published status, plan, and region are unverified.
PostgreSQL-connected through	Store application data such as products sales, orders, batches, and stock history	Provider, plan, and region aren't identified in the application configuration.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Seq2SeqNB LSTM, seed 0 — version not recorded (deep_seed0.pt) LightGBM tree forecaster — version not recorded (tree_mean.txt) Tree spread calibration (tree_spread.json) — not a standalone AI model	Self-developed (Trained on Runpod)	Self-trained	N/A

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic E-commerce Sales Data	Generated via Anthropic Claude	Synthetic	N/A	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

■ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit	Full-stack application development and live hosting.	Major
RunPod	Training Model	Major
Claude	Generating Synthetic Dataset	Moderate
Gemini	Generating vibe coding prompts for Replit.	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team took full responsibility for all AI-generated outputs, treating them as foundational drafts rather than final solutions. For the code generated via Replit, we manually reviewed the core logic—especially the FIFO batch tracking and database schemas, and conducted manual testing functions using our simulated barcode scanner and adjustable calendar. For Claude's synthetic dataset, we performed sanity checks to ensure it mirrored realistic e-commerce trends specifically in Malaysia without containing sensitive data. Ultimately, we own the final application's architecture, accuracy, and performance, ensuring all AI contributions meet our rigorous project standards.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

■ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Credits & Platform	Used as the primary cloud IDE for full-stack development, utilizing Replit Agent for UI/backend code generation and live application hosting.
RunPod	Platform & Credits	Used for training the PyTorch machine learning models (ensemble models) by leveraging high-performance cloud GPU compute resources.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

While highly beneficial for rapid development, the sponsor resources could technically be replaced. Replit provided seamless live-hosting and fast iteration, but we could substitute it with local IDEs (like VS Code) and local servers. RunPod offered crucial GPU acceleration to train our PyTorch ensemble models quickly within the hackathon's time limit, but this could be replaced by other cloud GPU providers or local hardware. Therefore, they were essential for our speed and efficiency during the event, but the core architecture itself is platform-agnostic.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

SImpan AI will collect a store name, username, password, and business records including product names, barcodes, categories, inventory levels, sales history, resupply orders, and stock activity. Passwords are stored as salted scrypt hashes. Session tokens are stored as hashed and sent in HTTP-only, Same Site cookies, marked Secure in production. Forecasting data is processed by a local Python service and is not sent to a third-party AI provider. There is no automatic data-retention period documented, records persist until deleted, and deleting a product also deletes its sales history.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The primary limitation is the 'cold start' problem: the model highly relies on historical transaction data, making predictions less accurate for newly launched products with zero sales history. Additionally, the model may struggle to predict sudden, viral demand spikes driven by external factors (e.g., social media trends), which is not captured in our dataset.

To mitigate these risks, we implemented two key features. First, a safety line function is added to relieve the stress if a sudden demand strike happens. Second, a 'Manual Purchase' override is built, ensuring merchants always retain final decision-making authority to ignore the AI and restock based on their own market intuition. This human-in-the-loop approach minimizes the risk of AI-driven overstocking or stockouts.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [ ] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [ ] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [ ] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [ ] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [ ] Any use of official SDC sponsor resources has been declared in Section 5.
- [ ] The prototype described here is accessible to judges through the links provided in Section 1.
- [ ] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



24/10/2026